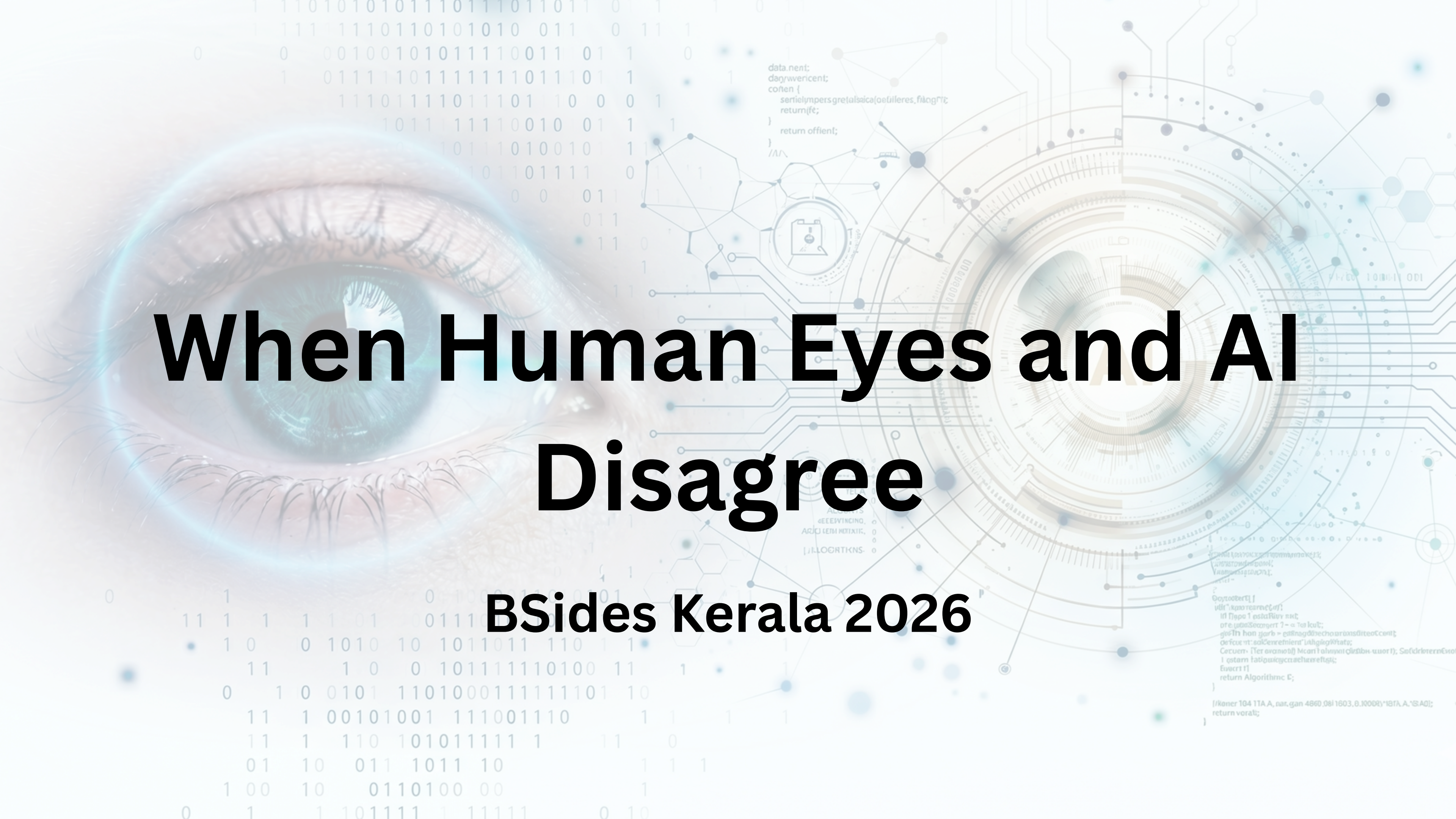




When Human Eyes and AI Disagree

BSides Kerala 2026

whoami 🤔

Sahil Gupta

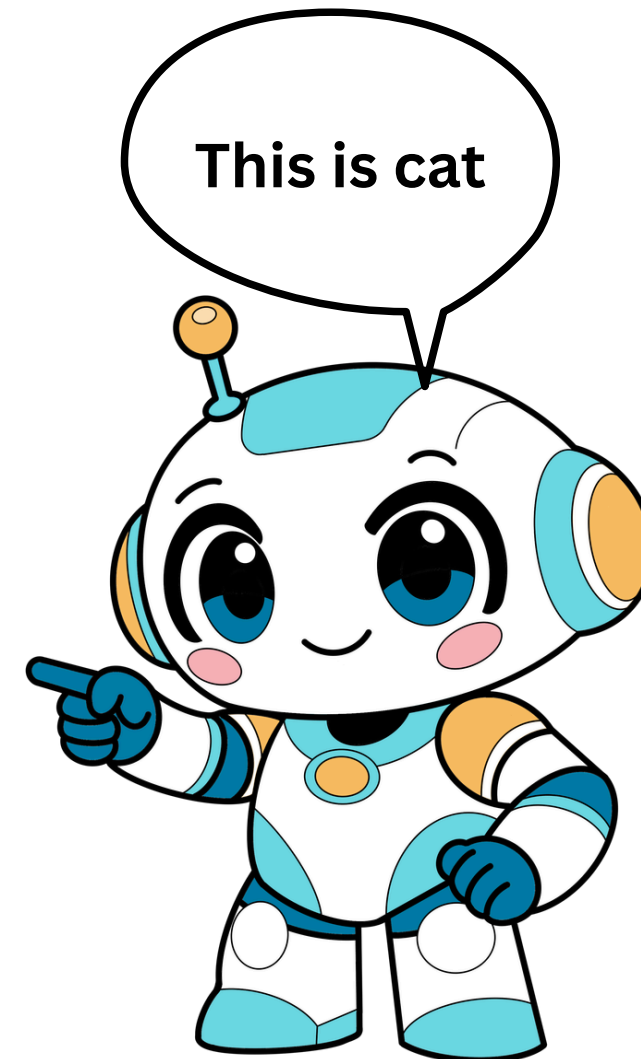
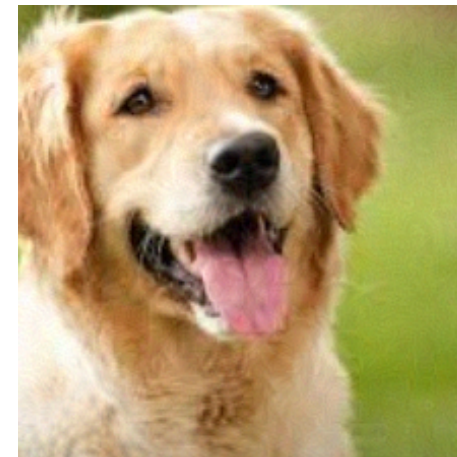
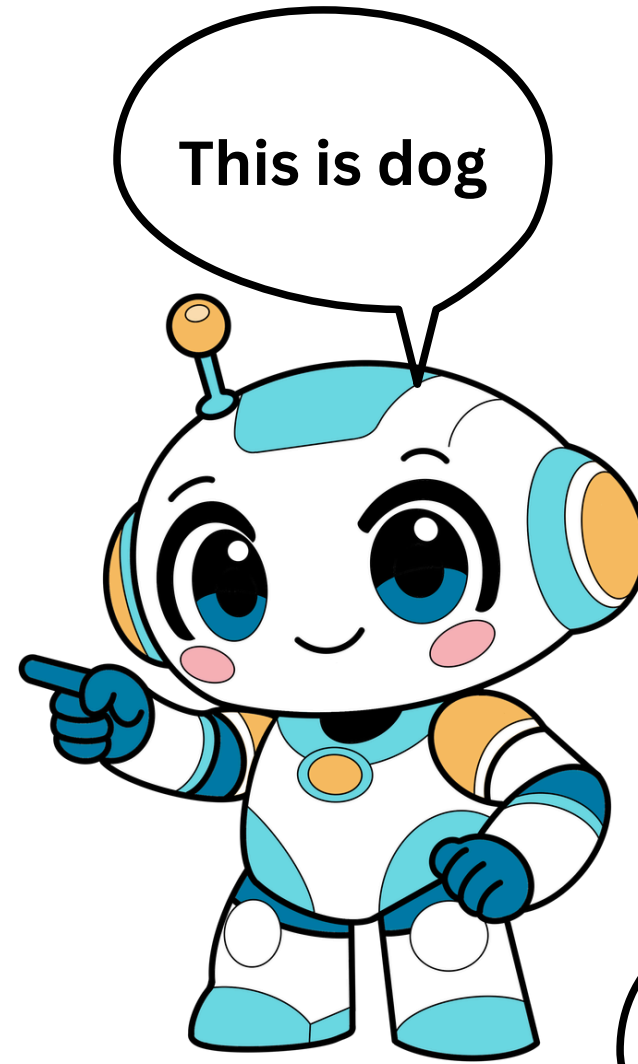
💻 Product Security @ Siemens Digital Industries Software

💻 Building and Scaling Product Security Program – Code to Cloud

🧐 Geeking out on AI for Security & Security for AI

🐶 Dog Owner

🌍 Fun life === Travel + New Experience + Food + Night Life

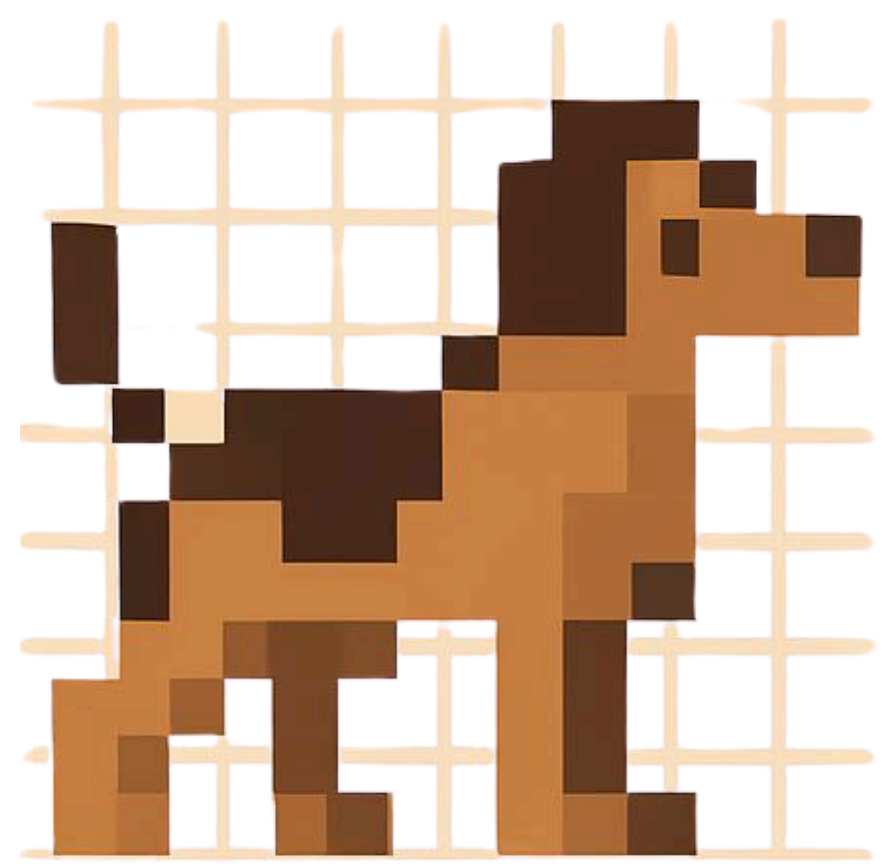


[R, G, B]
[210, 180, 140]

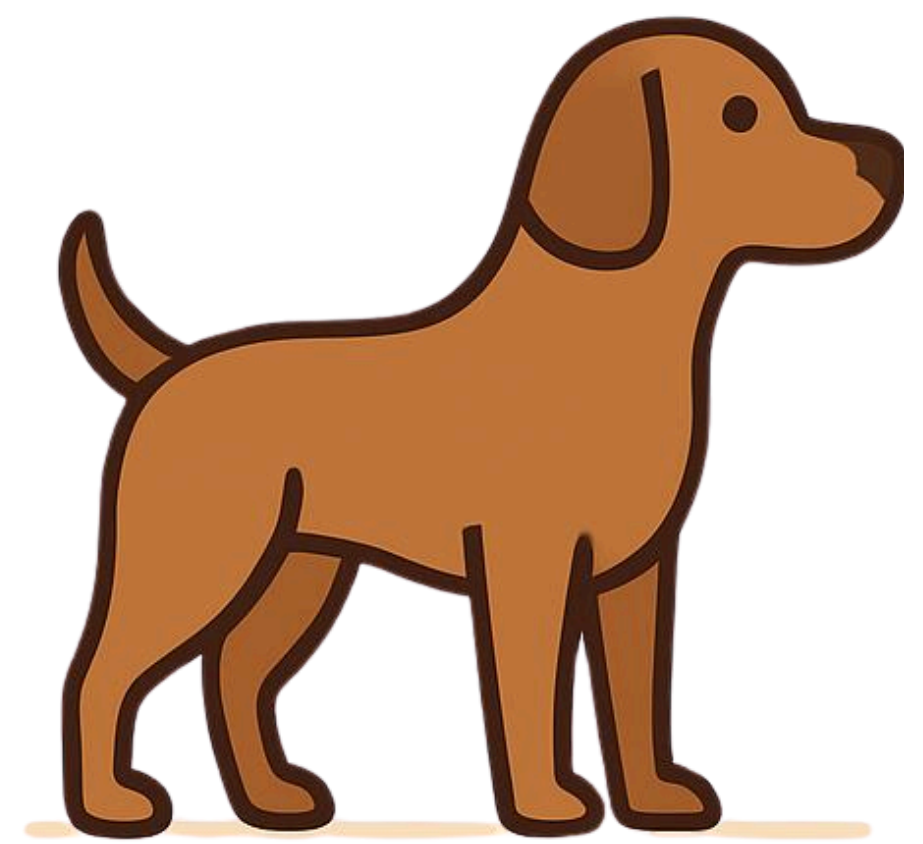
DETECTED: [EDGES,
COLOR GRADIENTS]

CLASS_ID:
'GOLDEN_RETRIEVE
R' (CONFIDENCE:
98.7%)

AI



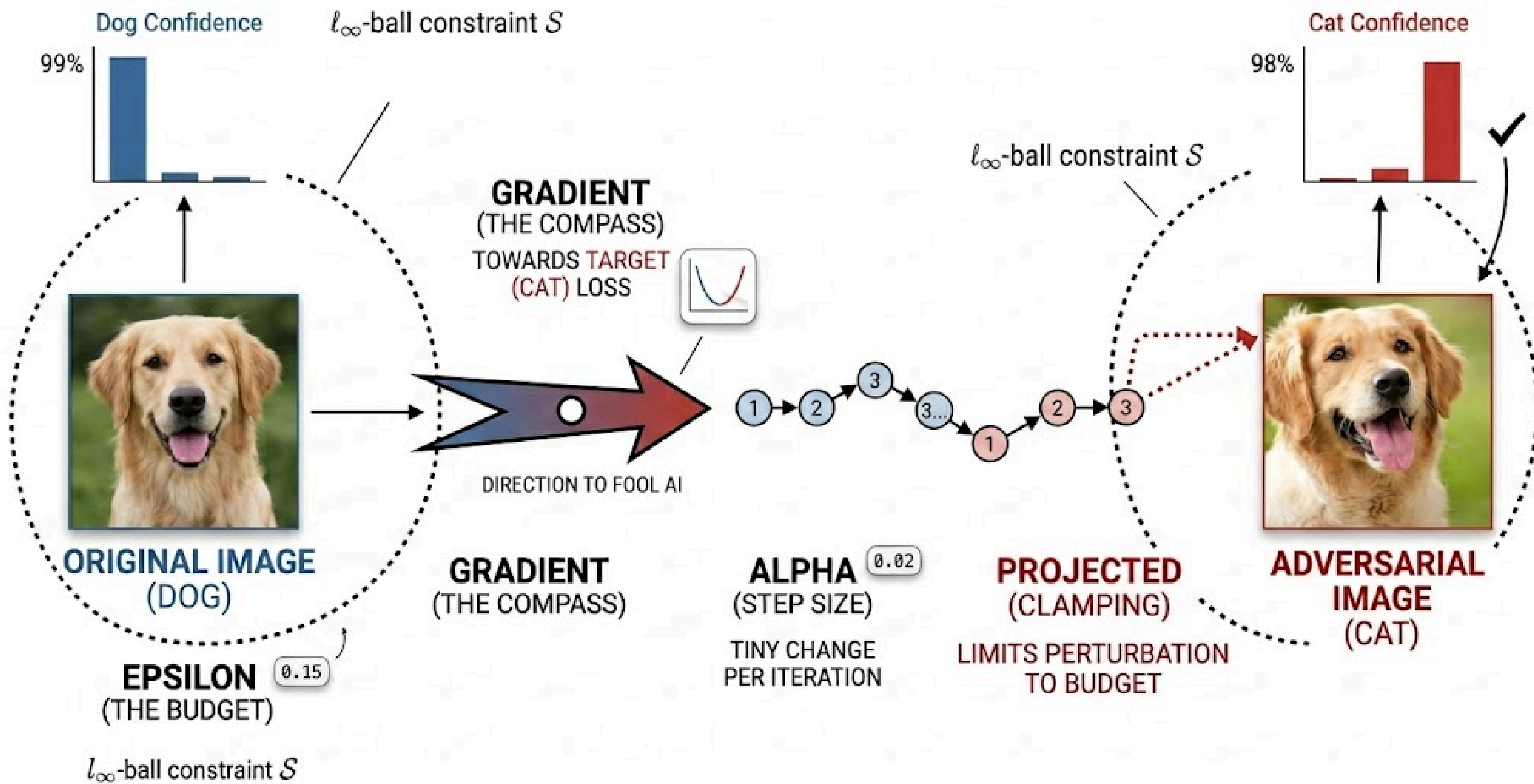
HUMAN

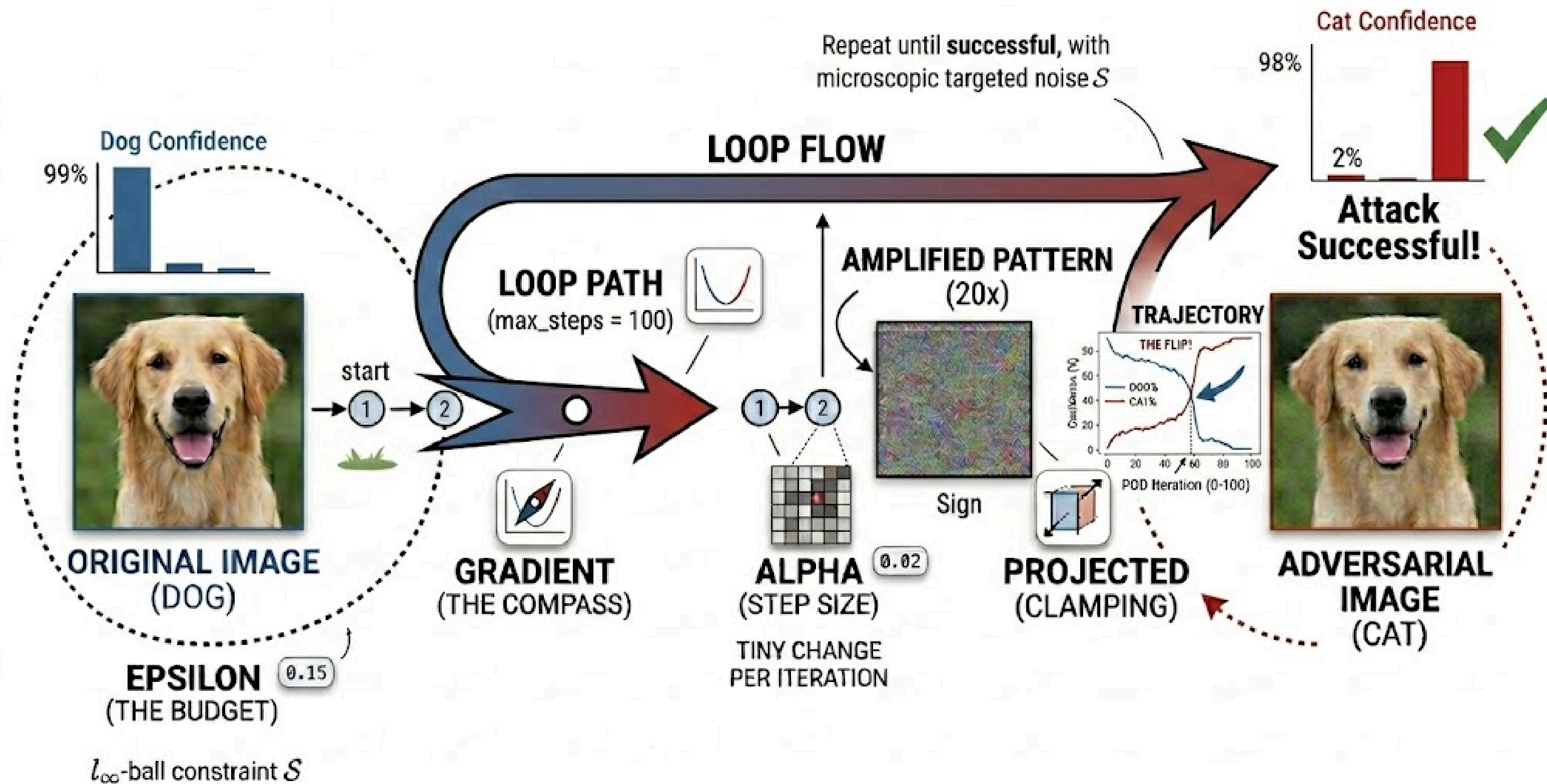


Understand Context

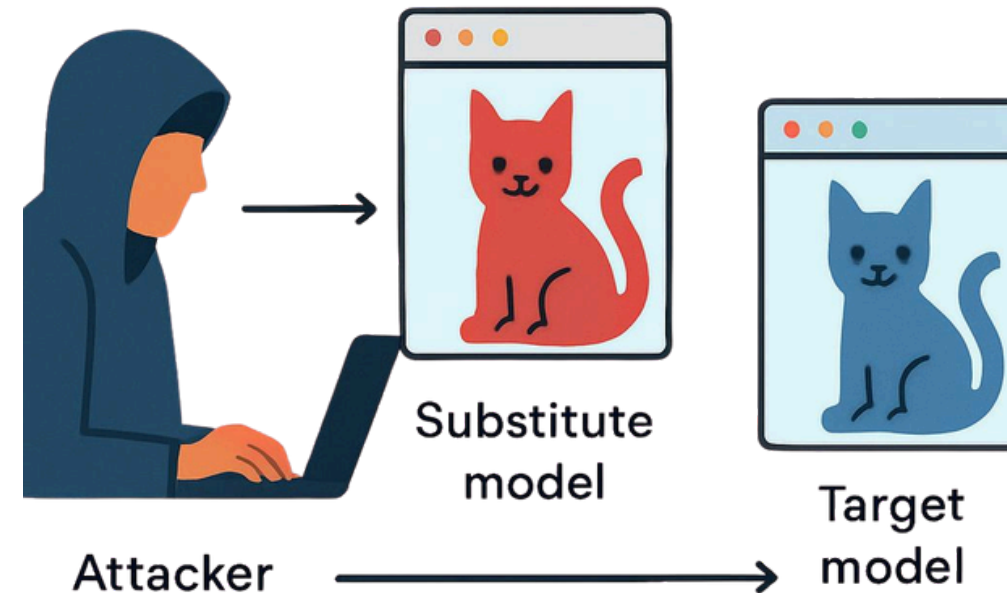
Connecting ideas

See the big picture

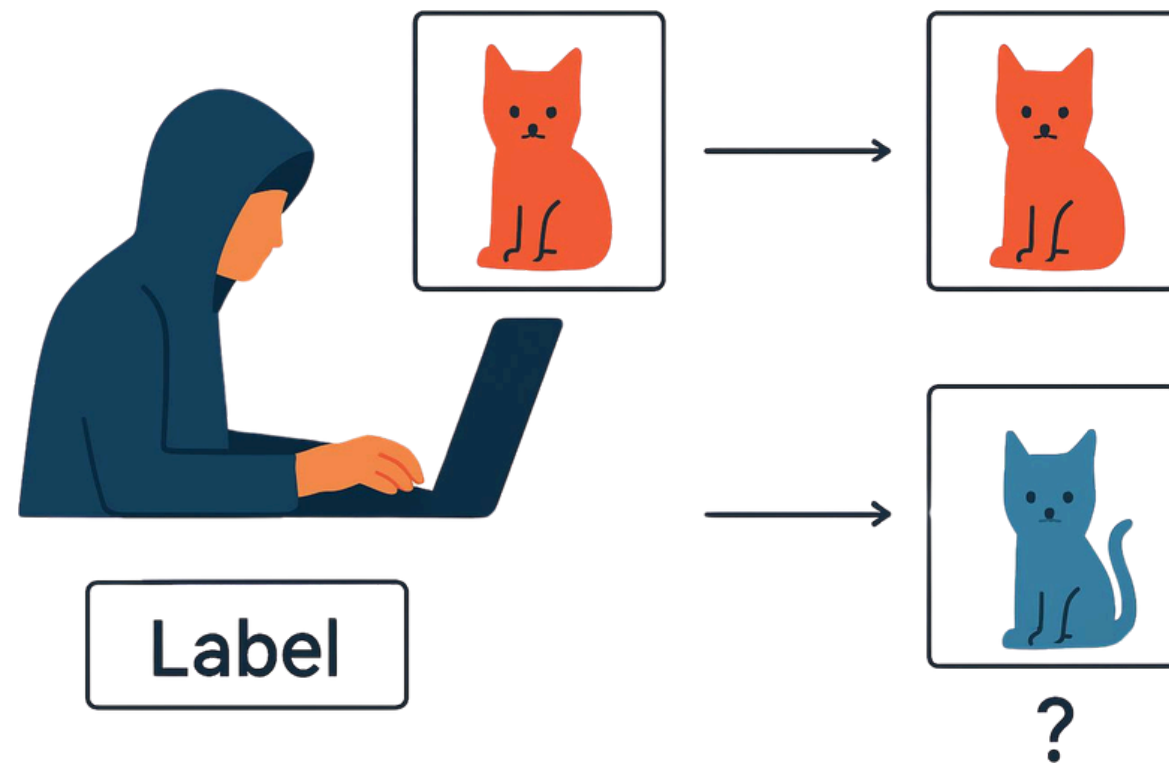




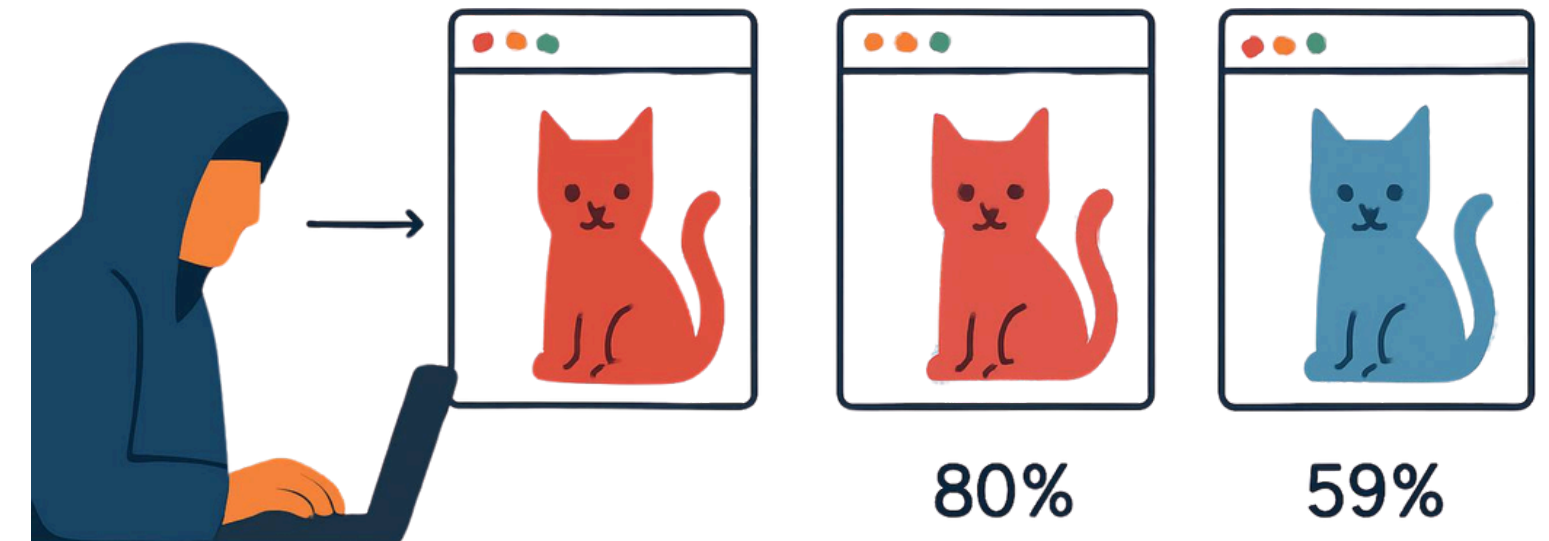
Black Box Attacks



Transfer-based Attack



Decision-based Query Attack

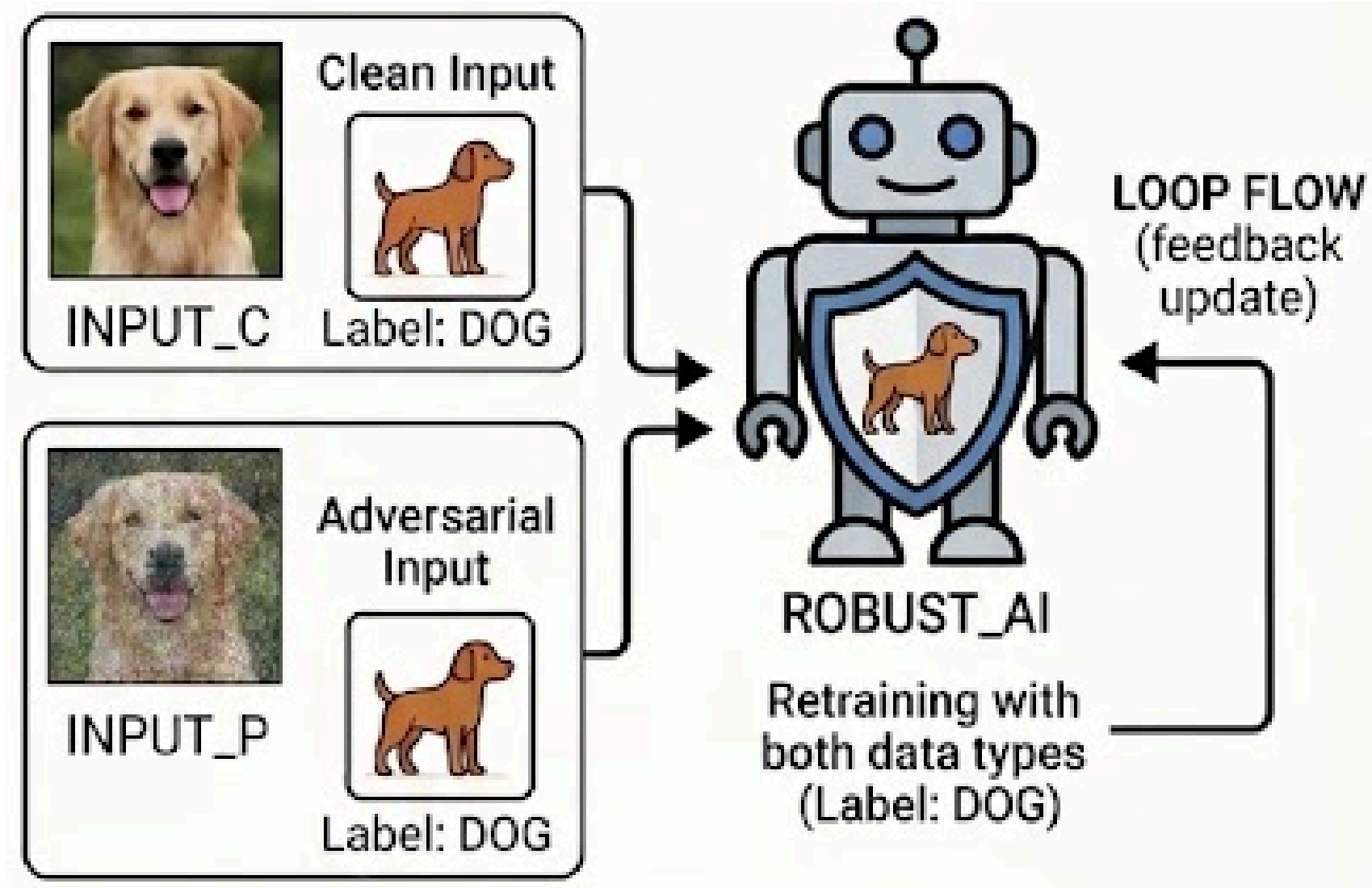


Score-based Query Attack

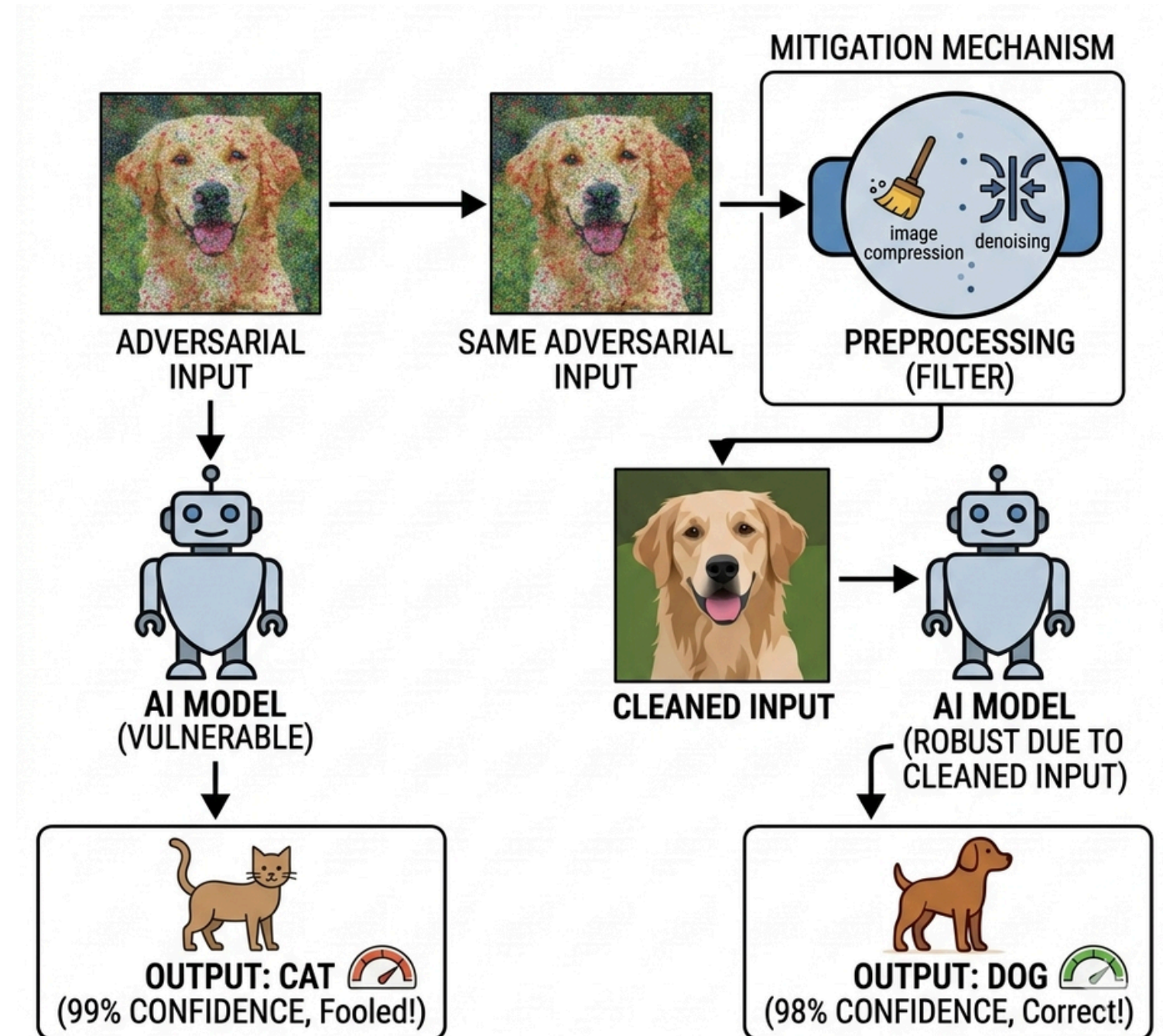
White Box Attacks

Attack Method	Speed	Primary Goal	Core Mechanism
Fast Gradient Sign Method (FGSM)	Very Fast	Untargeted	Single-step modification in the direction of the gradient's sign.
Projected Gradient Descent (PGD)	Moderate	Untargeted/Targeted	Iterative multi-step FGSM controlled steps, adjusting the direction each time.
Carlini & Wagner (C&W)	Very Slow	Targeted	Focused on optimization, balancing minimal changes with successful misclassification.
DeepFool	Moderate	Untargeted	Iteratively projects the input toward the closest decision boundary.
Jacobian-based Saliency Map Attack (JSMA)	Slow	Targeted	Alters very few pixels using a saliency map to maximize target class probability.

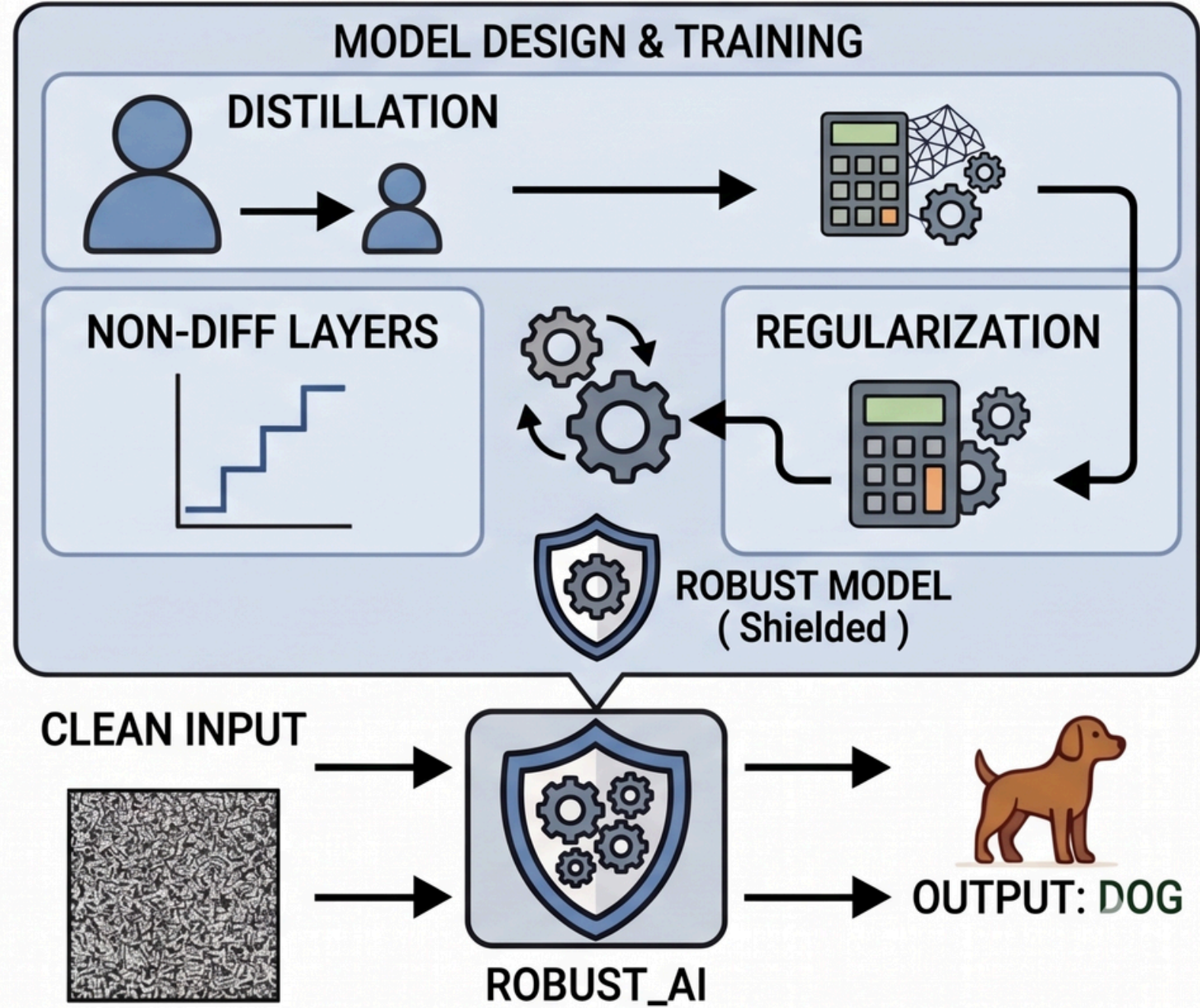
Adversarial Training



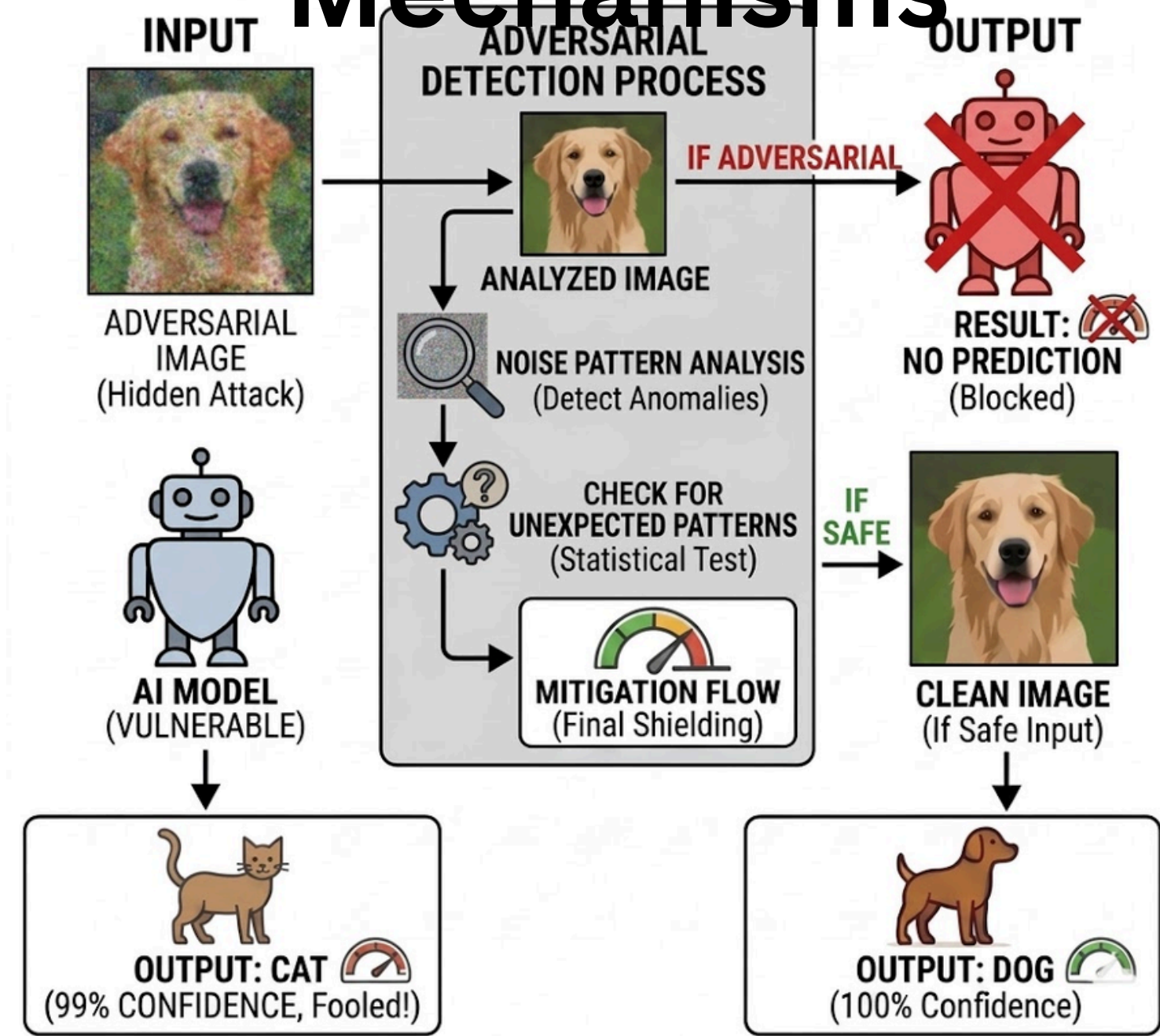
Input Preprocessing



Model Modifications



Detection Mechanisms



DEMO





